graphDataHistorian

Service Description

Service ID: *"graphDataHistorian"*

Abstract
This document describes the `graphDataHistorian` service, version 1.0.0, produced by the GraphDBManager support system of Arrowhead 5.2.2.

Contents

1 Overview **3**
1.1 Significant Prior Art . . . 3
1.2 How This Service Is Meant to Be Used . . . 3
1.3 Important Delimitations and Dependencies . . . 3

2 Service Interface **4**
2.1 operation list-systems . . . 4
2.2 operation list-services . . . 4
2.3 operation get-data . . . 4
2.4 operation put-data . . . 4

3 Information Model **5**
3.1 struct SenMLRecord . . . 5
3.2 struct ServiceListResponse . . . 5
3.3 struct SystemListResponse . . . 5
3.4 Primitives . . . 5

4 References **7**

5 Revision History **8**
5.1 Amendments . . . 8
5.2 Quality Assurance . . . 8

1 Overview

This document describes the `graphDataHistorian` service, which provides persistent storage and query of SenML time series per system and service. It is produced by the GraphDBManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Copy of the DataManager `dataHistorian` service (5.2.2) under its own service definition; same operations and data model, persisted as RDF in GraphDB [3].

1.2 How This Service Is Meant to Be Used

Producers append records with `put-data`; consumers query with `get-data`, optionally limited by `from`, `to`, `count`, `signals` and `signalCount`.

1.3 Important Delimitations and Dependencies

Data are stored as RDF in GraphDBManager's own GraphDB repository (see the SysDD for the vocabulary), without retention policy and without inference. With `service.authorization.enabled=true` every call is checked with ConsumerAuthorization.

2 Service Interface

This section describes the interfaces to the `graphDataHistorian` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

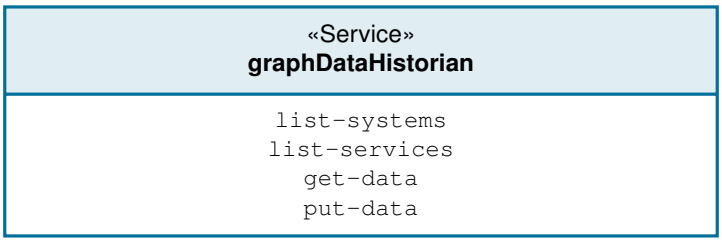


Figure 1: SysML-style block of the `graphDataHistorian` service and its operations.

The following interface operations are available.

2.1 operation list-systems () : SystemListResponse

Returns the systems with stored data.

2.2 operation list-services (Name) : ServiceListResponse

Returns the services of a system with stored data.

2.3 operation get-data (Name) : List<SenMLRecord>

Returns stored SenML records of the system and service, newest first, limited by the query parameters.

2.4 operation put-data (List<SenMLRecord>)

Appends SenML records for the system and service.

3 Information Model

Here, all data objects that can be part of the `graphDataHistorian` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.4, which are used to represent things like hashes and identifiers.

3.1 struct SenMLRecord

A SenML record (RFC 8428); a message is a JSON array of records.

Field	Type	Description
bn	String	SenML base name.
bt	String	SenML base time (seconds).
bu	String	SenML base unit.
bv	String	SenML base value.
bs	String	SenML base sum.
bver	String	SenML base version.
n	String	SenML name.
u	String	SenML unit.
v	String	SenML numeric value.
vs	String	SenML string value.
vb	Boolean	SenML Boolean value.
vd	String	SenML data value (Base64).
s	String	SenML sum.
t	String	SenML time (seconds, relative to the base time).
ut	String	SenML update time (seconds).

3.2 struct ServiceListResponse

Service names with data of a system.

Field	Type	Description
services	List<String>	Service names with stored data.

3.3 struct SystemListResponse

List of systems.

Field	Type	Description
systems	List<String>	System names with stored data.

3.4 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
String	A chain of characters.

4 References

[1] J. Delsing, “GraphDBManager – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-GraphDBManager-java-spring/documentation/v5.2.2/GraphDBManager-v5.2.2-SysD.pdf.

[2] ——, “graphDataHistorian HTTP/TLS/JSON – Interface Design Description (GraphDBManager),” Eclipse Arrowhead, IDD 5.2.2, 2026, graphDBManager-v5.2.2-graphDataHistorian-HTTP-IDD.pdf.

[3] Ontotext, “GraphDB 10.8 Documentation,” https://graphdb.ontotext.com/documentation/10.8/, 2025.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-28	5.2.2	Document created from the corresponding DataManager document for the GraphDB-backed copy GraphDBManager.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	